Grade 7 Practice

Standard 7.RP.1

Computation

1. A car goes 1/2 mile in 1/4 hour. What is its speed in miles per hour? ___ **2**

2. 3/4 pound of cheese costs $3. What is the price per pound? ___ **$4**

3. 2/3 cup of juice is poured in 1/2 hour. How many cups per hour? ___ **4/3**

4. 1 1/2 miles is walked in 1/2 hour. What is the speed in miles per hour? ___ **3**

Word problems

5. A snail moves 2/5 inch in 1/10 minute. How many inches per minute is that?
Answer: 4 inches per minute

6. Juice uses 3 1/2 cups for 7 people. How many cups per person?
Answer: 1/2 cup per person

Grade 7 Practice

Standard 7.RP.2

Computation

1. Is y = 3x a proportional relationship? Write yes or no. **yes**

2. In a table, x = 2 gives y = 6 and x = 5 gives y = 15. What is the constant of proportionc

3. Is y = x + 2 a proportional relationship? Write yes or no. **no**

4. Items cost $4 each. What is the constant of proportionality? ___ **4**

Word problems

5. Pay is $9 for 3 hours. What is the constant of proportionality in dollars per hour?
Answer: 3

6. A line passes through (0, 0) and (4, 10). What is the constant of proportionality?
Answer: 2.5

Grade 7 Practice

Standard 7.RP.3

Computation

1. What is 25% of 80? ___ **20**

2. What is a 15% tip on $40? ___ **6**

3. A $50 item is 20% off. What is the sale price? ___ **40**

4. A $200 amount increases by 10%. What is the new amount? ___ **220**

Word problems

5. A shirt costs $32. Sales tax is 8%. What is the total cost?
Answer: $34.56

6. A class has 30 students, and 40% are boys. How many boys are in the class?
Answer: 12 boys

Grade 7 Practice

Standard 7.NS.1

Computation

1. Solve: -3 + 8 = ___ **5**

2. Solve: 5 - (-2) = ___ **7**

3. Solve: -4 + (-6) = ___ **-10**

4. Solve: -7 - 2 = ___ **-9**

Word problems

5. The temperature is -5° and rises 9°. What is the new temperature?
Answer: 4°

6. An account balance is -$12. Then $20 is deposited. What is the new balance?
Answer: $8

Grade 7 Practice

Standard 7.NS.2

Computation

1. Solve: (-3)(4) = ___ **-12**

2. Solve: (-6)(-5) = ___ **30**

3. Solve: -24 ÷ 6 = ___ **-4**

4. Solve: (-1/2)(2/3) = ___ **-1/3**

Word problems

5. A diver goes down 5 meters every minute for 4 minutes. What is the total change in depth?
Answer: -20 meters

6. Temperature drops 3° each hour for 6 hours. What is the total change?
Answer: -18°

Grade 7 Practice

Standard 7.NS.3

Computation

1. Solve: -2.5 + 1.75 = ___ **-0.75**

2. Solve: 3/4 × (-4/9) = ___ **-1/3**

3. Solve: -1.2 × (-5) = ___ **6**

4. Solve: -1/2 + (-1/4) = ___ **-3/4**

Word problems

5. Ana's score is -2.5 points. She then gains 4.75 points. What is her new score?
Answer: 2.25 points

6. A diver is at -12.5 meters and rises 3.75 meters. What is the new depth?
Answer: -8.75 meters

Grade 7 Practice

Standard 7.EE.1

Computation

1. Expand: 3(x + 4) = ___ **3x + 12**

2. Simplify: 2x + 3x - x = ___ **4x**

3. Expand: -2(y - 5) = ___ **-2y + 10**

4. Factor: 6x + 9 = ___ **3(2x + 3)**

Word problems

5. Simplify: 5n + 2 + 3n - 1
Answer: 8n + 1

6. Simplify: 7 - 2(a + 3)
Answer: 1 - 2a

Grade 7 Practice

Standard 7.EE.2

Computation

1. Is 4(x + 3) equivalent to 4x + 12? Write yes or no. **yes**

2. Is 2x + 5 + x equivalent to 3x + 5? Write yes or no. **yes**

3. Is 3(x - 2) equivalent to 3x - 2? Write yes or no. **no**

4. Write 5x - 10 in factored form. ___ **5(x - 2)**

Word problems

5. Simplify 0.5x + 0.25x.
Answer: 0.75x

6. An item priced at p is 20% off. Write an expression for the sale price.
Answer: 0.8p

Grade 7 Practice

Standard 7.EE.3

Computation

1. Solve: 2x + 5 = 17. x = ___ **6**

2. Solve: 3(x - 4) = 9. x = ___ **7**

3. Solve: x/4 + 2 = 6. x = ___ **16**

4. Solve: -2x + 3 = 11. x = ___ **-4**

Word problems

5. A phone plan costs $15 plus $0.10 per text. Your total bill is $25. How many texts did you send?
Answer: 100 texts

6. Sam earns $9 per hour and pays an $8 fee. He nets $46. How many hours did he work?
Answer: 6 hours

Grade 7 Practice

Standard 7.EE.4

Computation

1. Solve: $x + 4 > 9$. $x >$ ___ **5**

2. Solve: $2x < 10$. $x <$ ___ **5**

3. Solve: $-3x \geq 12$. $x \leq$ ___ **-4**

4. Solve: $4(x + 1) = 20$. $x =$ ___ **4**

Word problems

5. A bike costs at most $150. Sam pays $40 down and then $25 each month. What is the greatest number of whole months he can pay?
Answer: 4 months

6. Two numbers add up to 18. The second is 2 more than the first. What is the first number?
Answer: 8

Grade 7 Practice

Standard 7.G.1

Computation

1. On a map, 1 cm = 5 m. What is 4 cm in meters? ___ **20**

2. A drawing has a scale of 1:100. A 6 cm line is how many cm in real life? ___ **600**

3. A scale is 2 inches = 10 feet. What is 3 inches in feet? ___ **15**

4. A scale is 1 cm : 2 km. What is 5 cm in km? ___ **10**

Word problems

5. A floor plan uses 1 inch for 3 feet. A room is 4 inches on the plan. How long is the real room?
Answer: 12 feet

6. A model car is built at a 1:24 scale. The real car is 96 inches long. How long is the model?
Answer: 4 inches

Grade 7 Practice

Standard 7.G.2

Computation

1. Can sides 3, 4, and 5 form a triangle? Write yes or no. **yes**

2. Can sides 2, 3, and 6 form a triangle? Write yes or no. **no**

3. A triangle has three 60° angles. What type of triangle is it? **equilateral**

4. A triangle has angles 90° and 40°. What is the third angle? ___ **50°**

Word problems

5. A triangle has angles of 30° and 70°. What is the third angle?
Answer: 80°

6. Two sides of a triangle are 5 and 7. Can the third side be 1? Write yes or no.
Answer: no

Grade 7 Practice

Standard 7.G.3

Computation

1. A cut parallel to the base of a cube makes what shape? ___ **square**

2. A cylinder is sliced straight down the middle. What shape is the cross section? ___

3. A sphere is sliced in any direction. What shape is the cross section? ___ **circle**

4. A triangular prism is cut parallel to its base. What shape is the cross section? ___ **tr**

Word problems

5. A cone is sliced horizontally. What shape is the cross section?
Answer: circle

6. A cylinder is sliced straight down the middle. What shape is the cross section?
Answer: rectangle

Grade 7 Practice

Standard 7.G.4

Computation

1. A circle has radius 3. Using π ≈ 3.14, find its area. ___ **28.26**

2. A circle has diameter 10. Using π ≈ 3.14, find its circumference. ___ **31.4**

3. A circle has radius 5. Using π ≈ 3.14, find its circumference. ___ **31.4**

4. A circle has diameter 4. Using π ≈ 3.14, find its area. ___ **12.56**

Word problems

5. A pizza has a radius of 7 inches. Using π ≈ 3.14, what is its area?
Answer: 153.86 square inches

6. A wheel has a diameter of 2 feet. Using π ≈ 3.14, what is its circumference?
Answer: 6.28 feet

Grade 7 Practice

Standard 7.G.5

Computation

1. What is the supplement of a 110° angle? ___ **70°**

2. What is the complement of a 35° angle? ___ **55°**

3. Vertical angles are equal. If one is 50°, what is the other? ___ **50°**

4. Two angles form a straight line. One is 65°. What is the other? ___ **115°**

Word problems

5. Two angles of a triangle are 50° and 60°. What is the third angle?
Answer: 70°

6. Angles 120° and x form a straight line. What is x?
Answer: 60°

Grade 7 Practice

Standard 7.G.6

Computation

1. A triangle has base 12 and height 5. What is its area? ___ **30**

2. A cube has side length 4. What is its volume? ___ **64**

3. A rectangular prism is 5 by 2 by 3. What is its volume? ___ **30**

4. A cube has side length 2. What is its surface area? ___ **24**

Word problems

5. A box is 3 feet by 4 feet by 5 feet. What is its volume in cubic feet?
Answer: 60 cubic feet

6. A wall is 10 feet by 8 feet. How many square feet of paint cover it?
Answer: 80 square feet

Grade 7 Practice

Standard 7.SP.1

Computation

1. Which sample is random? Write A or B. A) Ask 5 friends. B) Choose 20 names from a f

2. Is a sample chosen at random usually representative of the population? Write yes or

3. A survey uses only volunteers. Is this a random sample? Write yes or no. **no**

4. A sample is chosen by drawing names from a hat. Is it random? Write yes or no. **yes**

Word problems

5. Mia asks only the first 10 people who arrive at school. Is her sample random? Write yes or no.
Answer: no

6. Lee randomly picks 30 students from a group of 300. How many students are in the population?
Answer: 300 students

Grade 7 Practice

Standard 7.SP.2

Computation

1. A sample of 40 has 10 with a trait. Estimate the number in a population of 400. ___ **1**

2. In a sample of 10, 3 like red. Estimate for 500 people. ___ **150**

3. 25% of a sample of 200 students like math. About how many students like math? __

4. In a sample of 60, 15 say yes. What is the fraction of yes answers? ___ **1/4**

Word problems

5. A survey of 100 people shows 60% like pizza. About how many of 2,000 people like pizza?
Answer: 1,200 people

6. 8 of 20 students in a sample like the new lunch menu. About how many of 500 students like it?
Answer: 200 students

Grade 7 Practice

Standard 7.SP.3

Computation

1. Group A has a mean of 10 and group B has a mean of 12. Which mean is greater? Wri

2. Group A has a range of 5 and group B has a range of 9. Which group has more spre

3. Group A has a mean of 20 and group B has a mean of 15. What is the difference of th

4. Which has a greater spread: a range of 4 or a range of 10? Write the range. **10**

Word problems

5. Class A has a mean score of 80 and class B has a mean of 72. How many points higher is class A's mean?
Answer: 8 points

6. Class A has a range of 10 and class B has a range of 4. Which class has greater spread?
Answer: Class A

Grade 7 Practice

Standard 7.SP.4

Computation

1. Sample A has a mean of 70 and sample B has a mean of 65. What is the difference?

2. Sample A has a range of 12 and sample B has a range of 6. What is the difference in

3. Population A has a median of 40 and population B has a median of 35. Which has th

4. Means are 70 and 71, and each group has a range of 20. Is the difference meaningfu

Word problems

5. Two shops have mean prices of $12 and $15. What is the difference in the means?

Answer: $3

6. Group A has a median height of 150 cm and group B has a median of 160 cm. Which group is taller? Write A or B.

Answer: B

Grade 7 Practice

Standard 7.SP.5

Computation

1. What is the probability of rolling a 4 on a fair die? ___ **1/6**

2. What is the probability of flipping heads on a fair coin? ___ **1/2**

3. What is the probability of an impossible event? ___ **0**

4. What is the probability of a certain event? ___ **1**

Word problems

5. A bag has 10 marbles and 3 are red. What is the probability of picking a red marble?
Answer: 3/10

6. In the same bag, what is the probability of not picking a red marble?
Answer: 7/10

Grade 7 Practice

Standard 7.SP.6

Computation

1. A coin is flipped 100 times and lands heads 60 times. What is the relative frequency o

2. A spinner is spun 50 times and lands on blue 10 times. What is the relative frequency

3. A die is rolled 60 times and a six comes up 10 times. What is the relative frequency? _

4. An event happens 5 times in 20 trials. What is the estimated probability? ___ **0.25**

Word problems

5. A student tosses a cup 40 times and it lands upright 15 times. What is the relative frequency of landing upright?
Answer: 0.375

6. A relative frequency is 0.4 after 100 trials. How many successes were there?
Answer: 40

Grade 7 Practice

Standard 7.SP.7

Computation

1. A fair die is rolled. What is P(even)? ___ **1/2**

2. A model gives each of outcomes 1 to 6 equal probability. What is P(5)? ___ **1/6**

3. Two fair coins are flipped. What is P(two heads)? ___ **1/4**

4. A spinner has 8 equal sections. What is the probability of one section? ___ **1/8**

Word problems

5. A bag has 4 red and 6 blue marbles. What is the probability of picking blue?
Answer: 3/5

6. Three outcomes have probabilities 0.3, 0.5, and 0.1. If these are the only outcomes, what must the probability of a missing fourth outcome be? (Hint: the total must be 1.)
Answer: 0.1

Grade 7 Practice

Standard 7.SP.8

Computation

1. Two dice are rolled. What is P(sum is 7)? ___ **1/6**

2. Two coins are flipped. What is P(heads then tails)? ___ **1/4**

3. A card is drawn from a standard deck. What is P(red or ace)? ___ **7/13**

4. Two cards are drawn with replacement. What is P(both red)? ___ **1/4**

Word problems

5. A bag has 3 red and 2 blue marbles. A marble is picked, not replaced, then another. What is P(red then blue)?
Answer: 3/10

6. Two cards are drawn with replacement from a bag of 3 red and 2 blue cards. What is P(red then red)?
Answer: 9/25